

accordingly to control the car. Sensors and other components thus help achieve many tasks. This chapter discusses these parts and also details some projects.

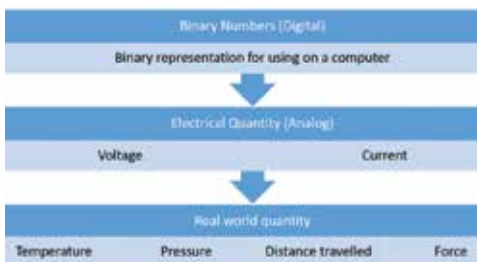
Components crucial to your project

Sensors and analog-to-digital converters

We work and live in the real world. But our devices can only deal with electrical quantity. Things get further complicated when we think of using computer-like devices to control the real world as they only understand numbers coded in binary language. However, computing makes tasks simple to achieve, so we achieve a nexus between the two by converting the electrical quantity to numbers for the computer to understand. By doing this, we're giving the computer an indirect way to understand things in the real world.

Sensors measure physical quantities (such as temperature) and convert them to an

electrical quantity (such as voltage or current), which in turn are converted into binary coded format by analog-to-digital converters for use on a computer.



Steps involved in making a computer analyze the real world

Actuators and digital-to-analog converters

'Actuators' are like sensors that work in reverse. They convert an electrical quantity (such as voltage and current) into real world quantity (such as heat or rotation to cause temperature change or physical motion).

To control actuators using a computer:

- ◆ We need to convert the computer's binary coded data (digital) into an electrical quantity (analog). We can accomplish this by using a 'Digital-to-Analog Converter' (commonly known as DAC). Digital-to-analog converters convert data from binary coded format on your computer into electrical quantity such as voltage or current.
- ◆ This electrical quantity drives the actuator to perform the required actions. An example of an actuator is a motor that converts electrical energy into rotational motion.